



PROGRAMMING FUNDAMENTALS

CS – 101

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HANDS ON PRACTICE WITH PYTHON

- print statement is used to print any value on screen
- Syntax: **print(some_value)**
- Examples:
 - `print (23 + 45)`
 - `print (34 – 3)`
 - `print (44 * 5)`
 - `print (395/57)`

EXERCISE

- If you run a 10 kilometer race in 43 minutes 30 seconds, what is your average time per mile? What is your average speed in miles per hour? (Hint: there are 1.61 kilometers in a mile)

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- Data:

Dist = 10km

Time = 43 min 30 sec

Avg time/mile (in seconds/mile) = ?

Avg time/mile (in minutes/mile) = ?

Avg Speed (in miles/hr) = ?

Solution:

Distance in km:

1 mile = 1.61km

1km = 1 mile/1.61

10 km = 10 /1.61

Time in seconds:

$43 \times 60 + 30$

Time in minutes:

$43 + 30/60$

Time in hours:

$(43 \times 60 + 30)/3600$

Avg time/mile (in seconds/mile) = $(43 \times 60 + 30)/(10/1.61)$

Avg time/mile (in minutes/mile) = $(43 + 30/60)/(10/1.61)$

Speed = Dist/Time

Speed (in miles/hr) = $\{10/1.61\}/\{(43 \times 60 + 30)/3600\}$

SOLUTION (CODING)

- Average time per mile (in seconds/mile):

```
print ( (minutes*60 + seconds) / (km/1.61) )
```

- Average time per mile (in minutes/mile):

```
print ( (minutes + seconds/60) / (km/1.61) )
```

- Average speed in miles/hr:

```
print ( (km/1.61) / ( (minutes*60 + seconds) / 3600) )
```

*** THE EXTRA SPACE THAT YOU SEE BETWEEN THE COMMANDS AND/OR BRACKETS IS **ONLY** TO MAKE IT READABLE FOR YOU, OTHERWISE THESE SPACES ARE **NOT** REQUIRED IN THE CODE ***